

Adilet Shynggys

Mechanical Engineering Student | Nuclear Systems & Fluid Mechanics

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Research Interests

- Capillary & Contact Line Dynamics
- Wetting / Non-Wetting Fluid Dynamics
- Advanced Reactor Design & Thermal-Hydraulics
- Nuclear Power Plant Accident Prevention

Skills

- **Computation:** MATLAB, Python, Origin-Pro, Mathematica
- **Standards:** ASME, GOST 33330-2015
- **CAD & Modeling:** SolidWorks (CSWA), Fluid Analysis
- **Fabrication:** Electrospinning, 3D Printing, Silicon Printing
- **Languages:** English (Fluent), Russian (Fluent), Kazakh (Native)

Awards & Honors

- Presidential Scholarship, Republic of Kazakhstan (2026)
- Nazarbayev University President's Scholarship
- Yessenov Scholarship (**Top 20** of 500+ applicants)
- Best Research Poster — B&R Initiative Conference (2025)
- Champion — NU AIChE Engineering Challenges
- 1st Place — Mechaton Engineering Case Competition
- Dean's List (4 consecutive semesters)

Leadership & Service

- **Charity Trail Run Organizer:** Managed logistics for a sports fundraiser; raised **600K KZT** (~\$1,300 USD) with 150+ runners.
- **Academic Physics Tutor:** Mentored 20+ freshman students, boosting average grades by **15%**.
- **HKU Muslim Society Volunteer:** Coordinated community outreach events during Ramadan in Hong Kong.
- **Mereke CF Volunteer:** Organized distribution of aid packages to vulnerable local families.
- **Cohort Representative:** Served as liaison between students and MAE Department faculty.

Education

Nazarbayev University

B.S. in Mechanical & Aerospace Engineering

GPA: 3.86 / 4.00 (**Ranked 1st in Cohort**) • Dean's List

Expected June 2027

University of Hong Kong

International Exchange Program

Collaborated with Prof. N. Fang, Prof. W. Yan & Prof. H. Kazerooni

Spring 2026

Research Experience

Research Assistant — Interfacial Fluid Mechanics

Summer 2026 – Present

• **NU MAE Dept. (Prof. Amgad Salama)**

- Investigating capillary dynamics and contact line behavior of wetting/non-wetting fluids in cylindrical and tapered geometries.
- Formulating mathematical models for dynamic contact angles, film deposition, and meniscus velocity profiles in multi-phase flows.

Research Assistant — Nanogenerators (TENG)

Jan – Dec 2025

• **NU Energy Conversion Lab**

- Fabricated Triboelectric Nanogenerators via electrospinning and silicon printing, enhancing energy output and power density by **15–20%**.
- Co-authored research paper in *Small* (**IF = 11.8, Q1**); awarded Best Research Poster at Belt & Road Initiative Conference 2025.

Engineering Design Projects

Nuclear Plant Core & Safety Design

Capstone Project

• **Nazarbayev University**

- Executing **thermal-hydraulic modeling** of a nuclear reactor core focused on accident prevention and safety margins under passive decay heat removal.

Ergonomic Train-Driver Seat Design

Mechaton Case Competition

• **Sponsored by Wabtec Corporation**

- Designed a high-durability driver seat compliant with **GOST 33330-2015** standards under 12-hour static and dynamic fatigue loading.

Multi-Stage Transmission Gearbox

HKU Exchange Project

- Sized shafts against **ASME** combined bending-torsional stress equations; selected keyways, bearings, and modules for target fatigue life.

Industry Experience

Product Support Engineering Intern

Jun – Jul 2025

• **Wabtec Corporation**

- Analyzed thermal performance of traction motors; updated technical drawings and process workflows, cutting material waste by **10%**.